

2 Optimal control of second-order systems

The optimal control problem for second-order systems, under sufficient differentiability conditions can be treated exactly as in Eq. (3). There, the augmented running cost featured inside the integral was a function on $T^*\mathcal{M} \oplus_{\mathcal{M}} \mathcal{E}$. In the second-order case, this naturally leads to an augmented running cost on $T^*T\mathcal{Q} \oplus_{T\mathcal{Q}} \mathcal{E}$. Assuming adapted local coordinates $(q, v, \lambda_q, \lambda_v)$ on $T^*T\mathcal{Q}$, and a generic controlled SODE $X = v \partial_q + f(q, v, u) \partial_v$, Eq. (3) transforms into

$$\begin{aligned} \tilde{J}_1(x, \lambda, u) &= \phi(q(T), v(T)) \\ &+ \int_0^T [C(q(t), v(t), u(t)) + \langle (\lambda_q(t), \lambda_v(t)), (\dot{q}(t) - v(t), \dot{v}(t) - f(q(t), v(t), u(t))) \rangle_{T\mathcal{Q}}] dt. \end{aligned} \quad (7)$$

We refer to this as the *first-order version* of the optimal control problem since the controlled SODE appears as a first order system.

The second-order constraint, i.e. $\dot{q} = v$, may be added implicitly, leading to a new augmented cost function

$$\tilde{J}_2(y, u) = \phi(q(T), \dot{q}(T)) + \int_0^T [C(q(t), \dot{q}(t), u(t)) + \kappa(t)^\top (\ddot{q}(t) - f(q(t), \dot{q}(t), u(t)))] dt. \quad (8)$$

Here, the curve $y = (q, \kappa)$ is a curve on $T^*\mathcal{Q}$. One must also understand u to denote the curve defined on the fibers along the tangent lift of q , i.e. $(q(t), \dot{q}(t), u(t)) \in \mathcal{E}$. The remaining constraint is now to be interpreted as a function on $T^{(2)}\mathcal{Q} \oplus_{T\mathcal{Q}} \mathcal{E}$.

Taking variations we find that the necessary conditions for optimality provided by $\tilde{J}_1$ are

- (state dynamics) $\dot{q}(t) = v(t)$,
 $\dot{v}(t) = f(q(t), v(t), u(t))$,
- (adjoint dynamics) $\dot{\lambda}_q(t)^\top = D_1 C(q(t), v(t), u(t)) - \lambda_v(t)^\top D_1 f(q(t), v(t), u(t))$,
 $\dot{\lambda}_v(t)^\top = D_2 C(q(t), v(t), u(t)) - \lambda_v(t)^\top D_2 f(q(t), v(t), u(t)) - \lambda_q(t)^\top$,
- (maximization) $0 = D_3 C(q(t), v(t), u(t)) - \lambda_v(t)^\top D_3 f(q(t), v(t), u(t))$,
- (transversality) $\lambda_q(T)^\top = -D_1 \phi(q(T), v(T))$,
 $\lambda_v(T)^\top = -D_2 \phi(q(T), v(T))$;

while those provided by $\tilde{J}_2$ are

- (state dynamics) $\ddot{q}(t) = f(q(t), \dot{q}(t), u(t))$,
- (adjoint dynamics) $\ddot{\kappa}(t)^\top = \frac{d}{dt} [D_2 C(q(t), \dot{q}(t), u(t)) - \kappa(t)^\top D_2 f(q(t), \dot{q}(t), u(t))]$
 $- D_1 C(q(t), \dot{q}(t), u(t)) + \kappa(t)^\top D_1 f(q(t), \dot{q}(t), u(t))$,
- (maximization) $0 = D_3 C(q(t), \dot{q}(t), u(t)) - \kappa(t)^\top D_3 f(q(t), \dot{q}(t), u(t))$,
- (transversality) $\kappa(T)^\top = -D_2 \phi(q(T), \dot{q}(T))$,
 $\dot{\kappa}(T)^\top = D_1 \phi(q(T), \dot{q}(T))$
 $+ D_2 C(q(T), \dot{q}(T), u(T)) + D_2 \phi(q(T), \dot{q}(T)) D_2 f(q(T), \dot{q}(T), u(T))$.

These, lead to the following easy to check

Theorem 2.1. *The necessary optimality conditions provided by $\tilde{J}_1$ and $\tilde{J}_2$ are equivalent under the identification $\lambda_v = \kappa$.*